Markov Chains (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.03

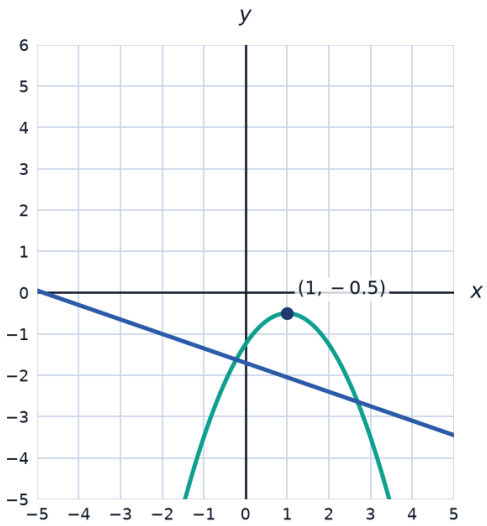
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Are $\mathbf{u} = (2; 0; 0)$ and $\mathbf{w} = (7; 0; 0)$ parallel? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the number of ways to arrange the 7 letters in the word SUCCESS.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.

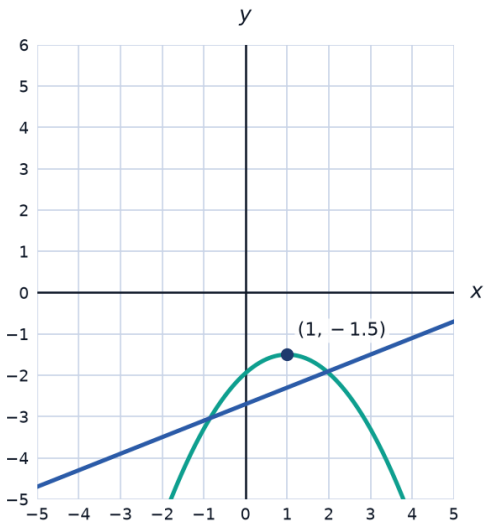
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

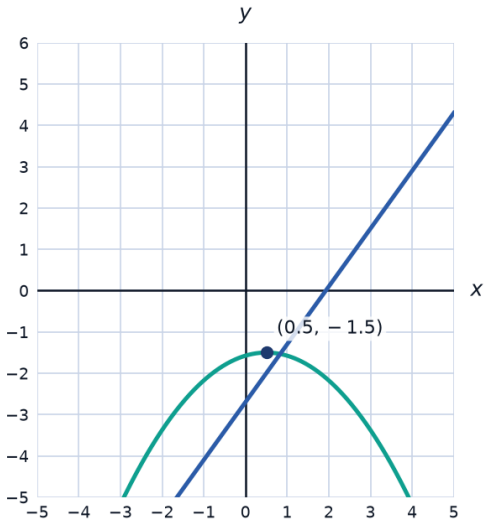
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

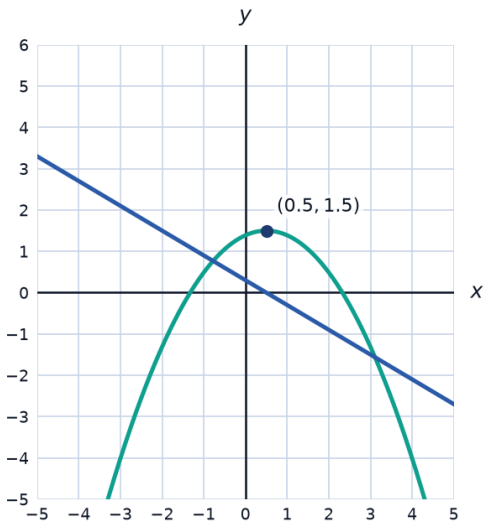
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Matrices (AI HL)**

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Proof by Mathematical Induction**

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

[About AS Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1

b. 0

c. The number of states

d. $\frac{1}{n}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. $[0.60, 0.40]$

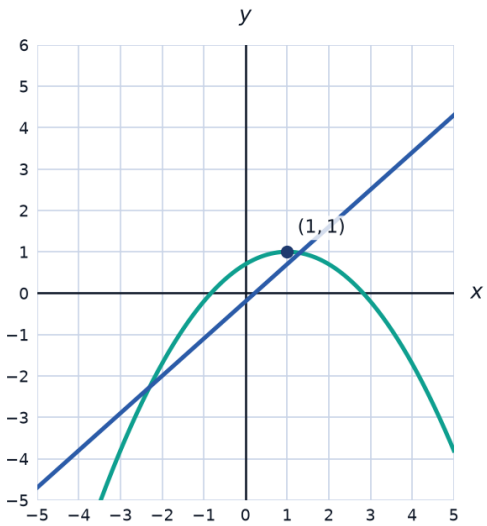
b. $[0.55, 0.45]$

c. $[0.66, 0.34]$

d. $[0.48, 0.28]$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the Markov property (memoryless property) in stochastic processes?



a. The future state depends only upon the current state, not on the sequence of states that preceded it

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does entry $(T^3)_{2,1}$ represent?

a. The probability of transitioning from State 2 to State 1 in exactly 3 steps

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi} T = \boldsymbol{\pi}$ and $\sum \pi_i = 1$

b. $T\boldsymbol{\pi} = \mathbf{0}$

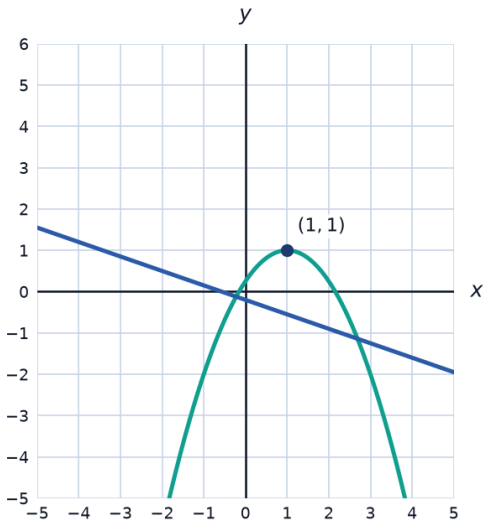
c. $\boldsymbol{\pi} T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.



a. $\boldsymbol{\pi} = [0.4, 0.6]$

b. $\boldsymbol{\pi} = [0.5, 0.5]$

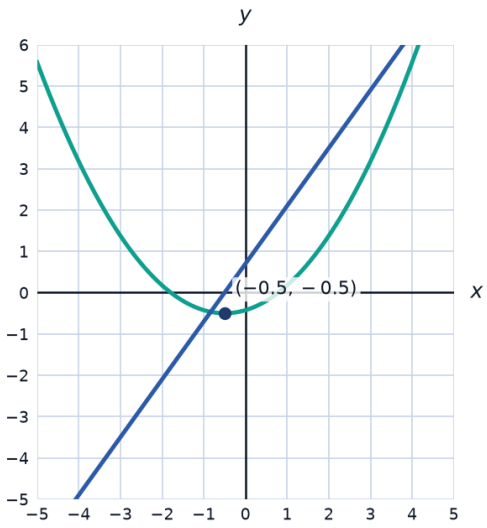
c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?



a. When the transition matrix is regular (some power T^k has strictly positive entries)

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is an absorbing state in a Markov chain?

a. A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$)

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. The expected number of visits to each transient state before absorption

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$

b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Triple integral $\iiint_V 1 \, dV$ gives:

a. volume of V

b. 0

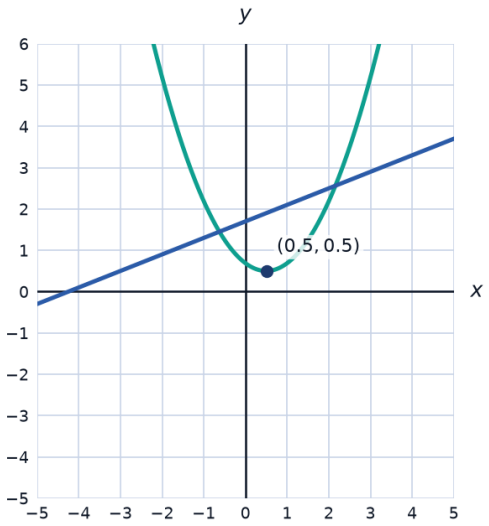
c. surface area

d. 1

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use polar coordinates to evaluate $\iint_R \sqrt{x^2 + y^2}\, dA$, where R is the annulus $1 \le x^2 + y^2 \le 4$.



a. $\frac{14}{3}$

b. $\frac{14\pi}{3}$

c. $\frac{28\pi}{3}$

d. $\frac{7\pi}{3}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Optimize area of rectangle with perimeter 20: max area $=$

a. 25

b. 20

c. 50

d. 100

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Riemann sum step width $\Delta x =$

a. $1/n$

b. $(b - a)/n$

c. $n(b - a)$

d. $(f(b) - f(a))/n$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Volume of cone $z = \sqrt{x^2 + y^2}$, $z \le 1$ in cylindrical:

a. $2\pi/3$

b. $\pi/3$

c. π

d. $\pi/2$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the critical point of $f(x) = x^2 - 4x + 3$.

a. 1

b. 3

c. 2

d. -2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Proof by Mathematical Induction

By the Intermediate Value Theorem, does $f(x) = x^3 - x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. No — f is not continuous on $[1, 2]$

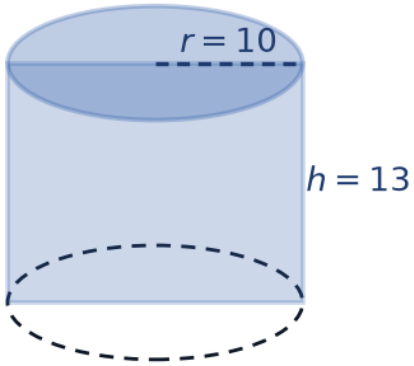
c. Yes — $f(1)$ and $f(2)$ have opposite signs

d. Cannot be determined without more information

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

The moment of inertia about the z-axis: $I_z = \iiint_V (x^2 + y^2)\rho\, dV$. For a uniform cylinder $r^2 \le R^2$, $0 \le z \le h$, $\rho = \text{const}$:



a. $\rho\pi R^2 h/2$

b. $\rho\pi R^2 h \cdot R^2/2$

c. $\pi R^2 h$

d. $\rho R^4 h$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Volume of spherical shell $1 \le \rho \le 2$:

a. 4π

b. $28\pi/3$

c. 8π

d. $4\pi/3$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.04 Partial Fractions

For a region symmetric about the y-axis, $\iint_D x\, dA =$

a. 0

b. -1

c. 1

d. $A(D)$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A rectangle has perimeter 40. Find the width w that maximises its area.

a. 10

b. 20

c. 5

d. $\frac{40}{3}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

$\int_0^1 \int_0^x y\,dy\,dx$: inner integral gives $x^2/2$, so outer integral $=$

a. 1

b. 1/4

c. 1/2

d. 1/6

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

First step in proving $\sqrt{3}$ is irrational by contradiction:

a. assume $\sqrt{3}$ is rational and never simplify

b. assume $\sqrt{3} = p/q$ in lowest terms

c. square both sides immediately

d. assume p, q share a common factor

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Moment of inertia about z-axis of uniform ball radius R, mass m:

a. $2mR^2/5$

b. $2mR^2/3$

c. $mR^2/2$

d. mR^2

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